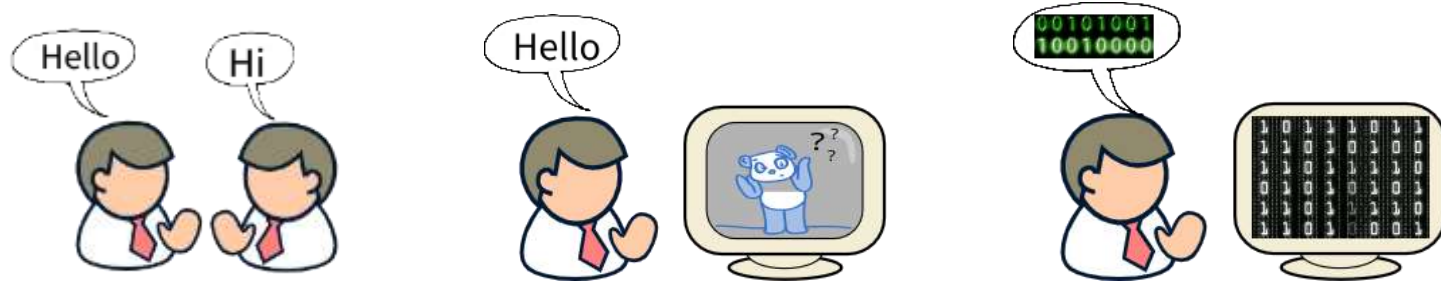


Introduction to Programming Logic & Flow Control

What is a program?



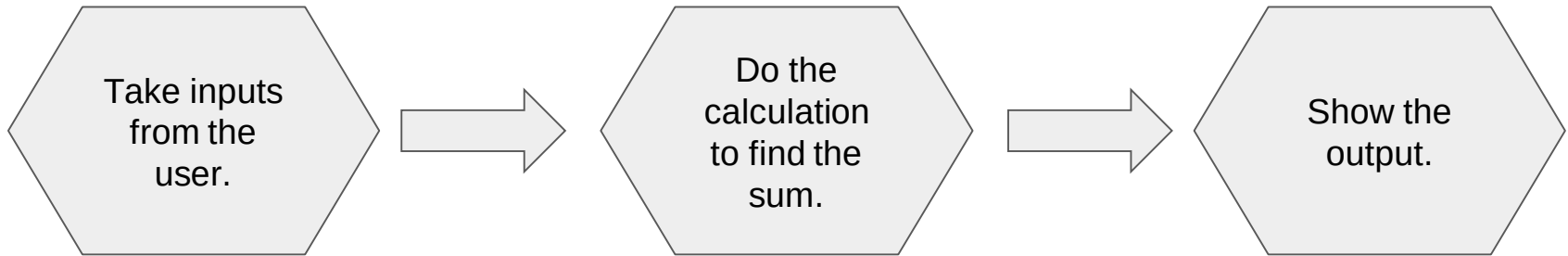
A computer program is the fundamental building block of software development, enabling computers to perform a variety of tasks to provide valuable functionality to users.

In other words, we use programs to communicate with computers in order to give specific instructions to it.

Programming Logic

- Systematic process of reasoning and formulating solutions to specific problems.
- Breaking down problems into smaller, manageable tasks and creating step-by-step solutions.

Example: You need to write a program to take two numbers from user and show their summation in the output. What would be the steps here?



Program Flow Control

Program flow control refers to the order in which instructions are executed within a program based on certain conditions or control structures.

- Sometimes, program flow can be sequential.
- However, most of the time program, flows are non-sequential.
- Program flow can loop back to the previous section of the program.
- It is also possible to skip or stop executing certain sections of the program based on some condition.

What is a Flowchart?

Visual representation of a program that uses various shapes, symbols and arrows to explain the sequence or the flow of events.

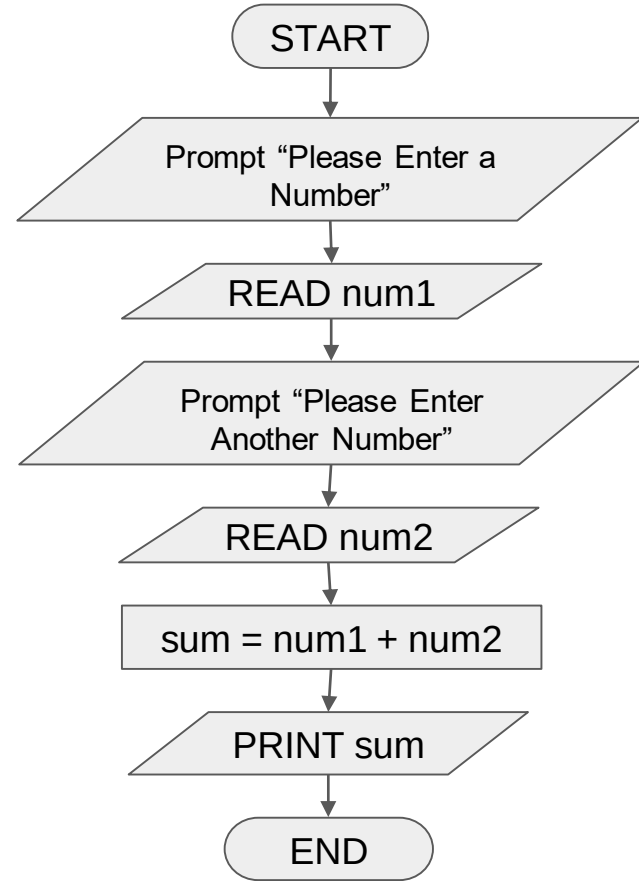
Few Basic shapes of a flowchart:

Shape	Name	Description
	Oval	Denotes the Start or the End of a program.
	Rectangle	Denotes any Calculation that the program must complete.
	Parallelogram	Denotes Prompt Messages or Input to the program or Output from the program.
	Circle	This is called the Connector because it is used to combine multiple blocks of code or divide them.
	Arrow	Illustrates the Flow of a program by merging from one shape to another.

Scenario to Flowchart

Let's go back to the previously discussed scenario and try to visualize it using a flowchart.

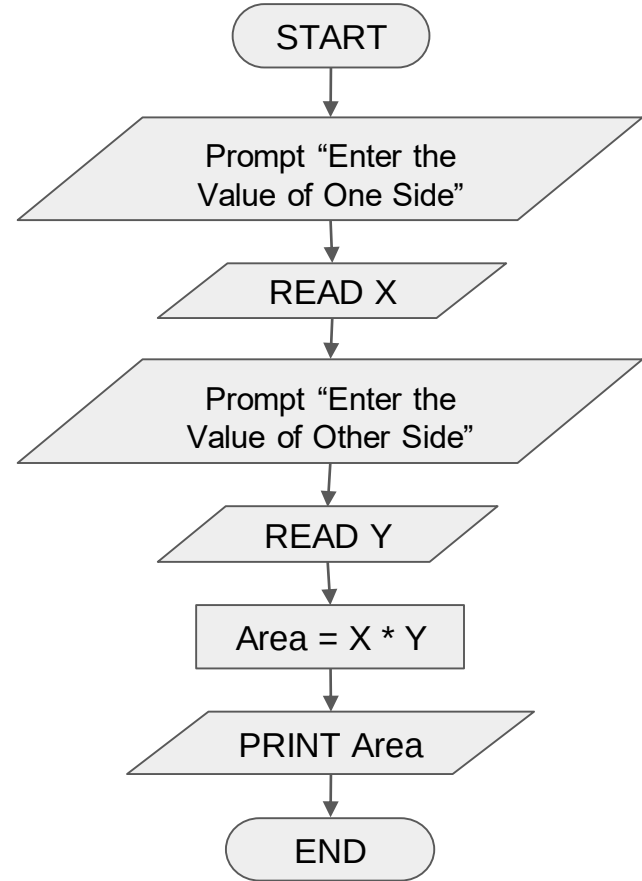
Scenario: Think of a program that takes two numbers from the user and shows their summation in the output. Represent this program using a flowchart.



Scenario to Flowchart (Contd)

CLASSWORK #1

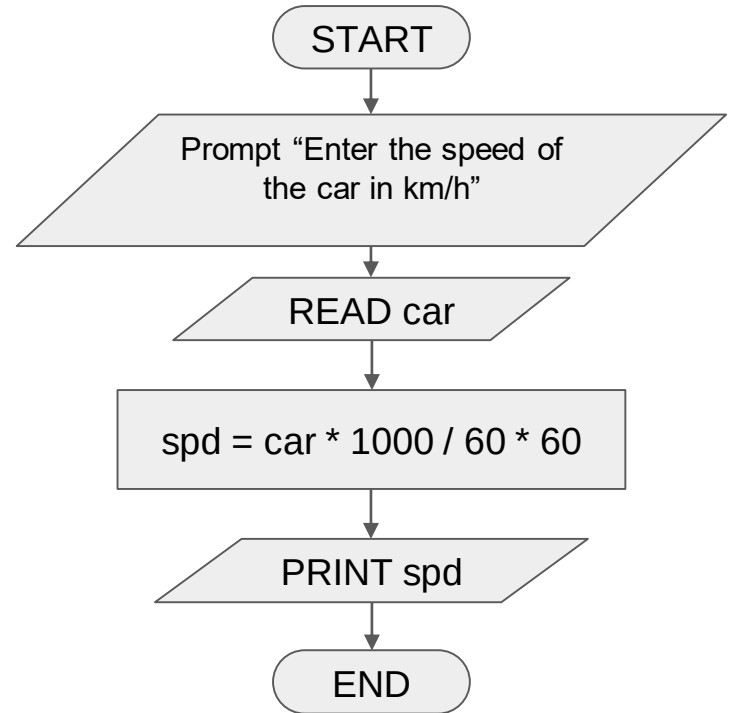
Scenario: Think of a program that calculates and shows the area of a rectangle after taking the value of its two sides as input. Represent this program using a flowchart.



Scenario to Flowchart (Contd)

CLASSWORK #2

Scenario: Think of a program that calculates and shows the speed of a car in meter/second after getting the speed as kilometer/hour from the user as input. Represent this program using a flowchart.

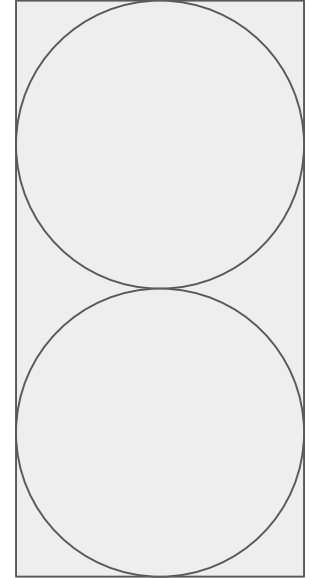


Scenario to Flowchart (Contd)

Visualization

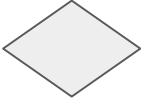
CLASSWORK #3

Scenario: Imagine a rectangle inside of which two circles with radius of R can fit perfectly. Each circle is touching three different sides of the rectangle. Given the value of radius as input, you must calculate and show the area inside the rectangle which is unoccupied by the circles. Represent this program using a flowchart.



Scenario to Flowchart (Contd)

What we saw in the previous slides was an example of sequential program flow. However, most of the cases program flow are non-sequential. In those scenarios, we use a **Diamond shape** to decide where the program flow will continue.

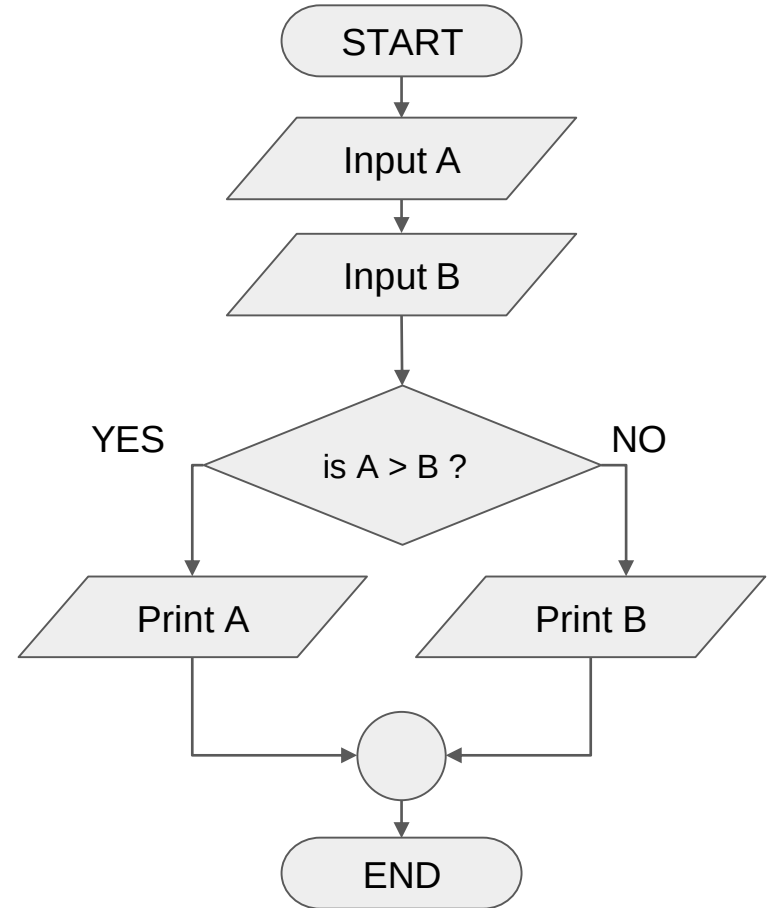
Shape	Name	Description
	Rhombus or Diamond	Denotes a Decision , based on which it is determined that which way the program flow will continue. Even though only one arrow flows into this shape but two arrows comes out of it.

Scenario to Flowchart (Contd)

Let's check out an example of non sequential program flow.

Scenario: Think of a program that takes two numbers from user and shows the largest number in the output. Represent this program using a flowchart.

Note: More detail on this in Chapter 7.



THANK YOU!